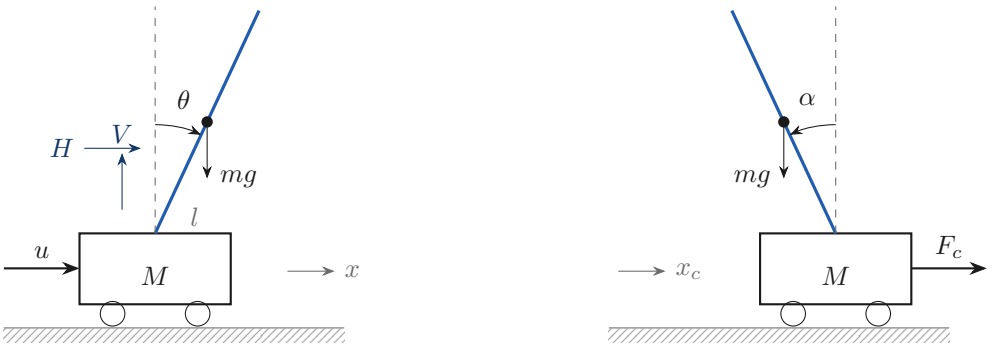
cart-pole model, solved

Step 1 of P&S Mission Impossible CartPole 4.0 | Ale, 01.10.2026

The whole model is two coupled equations. They come from Ogata's Example 3-5 (Modern Control Engineering, printed p. 68–69) with three changes the handout asks for: the angle is measured the other way ($\theta = -\alpha$), the cart gets friction $-B_f \dot{x}_c$, and the input is the motor voltage instead of a force. Everything the course does later is algebra on these two lines: the simulator integrates them, the friction fit (Step 2) rearranges them, and the controllers (Steps 3–5) use their linearisation.

Notation follows the lab: $m = m_p = 0.127\,\text{kg}$ pole, $M = m_c = 0.57\,\text{kg}$ cart, $l = r = 0.1778\,\text{m}$ pivot to centre of mass, I the pole's inertia about its centre of mass. The Ogata PDF renders θ as the letter "u"; read every "u" inside a sine or with dots on it as θ.

1 Cutting the pendulum at the pivot leaves four Newton equations



a **Ogata:** $\theta > 0$ **clockwise** **b** **Lab:** $\alpha > 0$ **counter-clockwise**

Figure 1. Both conventions put zero at upright. A positive Ogata angle tilts the pole toward $+x$; a positive lab angle tilts it toward $-x$. Panel a shows the pivot forces H, V acting on the rod, drawn beside the pivot for legibility; the cart feels the same forces reversed.

The centre of mass of the rod sits at

$$x_G = x + l\sin\theta, \qquad y_G = l\cos\theta.$$

Ogata cuts the system at the hinge. The hinge carries two forces nobody measures, a horizontal H and a vertical V. They push the rod one way and the cart the other, so they enter the rod's equations with $+$ and the cart's with $-$:

$$I\ddot{\theta} = Vl\sin\theta - Hl\cos\theta \qquad \text{rod rotating about its centre of mass} \qquad (3\text{-}9)$$

$$m\,\tfrac{d^2}{dt^2}(x + l\sin\theta) = H \qquad \text{rod's centre of mass, horizontal} \qquad (3\text{-}10)$$

$$m\,\tfrac{d^2}{dt^2}(l\cos\theta) = V - mg \qquad \text{rod's centre of mass, vertical} \qquad (3\text{-}11)$$

$$M\ddot{x} = u - H \qquad \text{cart, horizontal} \qquad (3\text{-}12)$$

In (3-9) each pivot force acts a distance l from the centre of mass, and its torque is force times perpendicular lever arm: V has lever $l\sin\theta$, H has $l\cos\theta$. The cart's vertical equation is not needed: the rail absorbs it. **The rest of the derivation only eliminates H and V.**

2 Small angles turn the four equations into Ogata's two

Near upright, $\sin\theta \approx \theta$, $\cos\theta \approx 1$ and $\theta\dot{\theta}^2 \approx 0$ (a product of small quantities). Then $l\cos\theta$ is constant, (3-11) gives $V = mg$, and the rest reads

$$I\ddot{\theta} = Vl\theta - Hl \tag{3-13}$$

$$m(\ddot{x} + l\ddot{\theta}) = H \tag{3-14}$$

$$0 = V - mg \tag{3-15}$$

Put H from (3-14) into (3-12), and put $V = mg$ and H into (3-13):

$$(M+m)\,\ddot{x} + ml\,\ddot{\theta} = u \tag{3-16}$$

$$(I + ml^2)\,\ddot{\theta} + ml\,\ddot{x} = mgl\,\theta \tag{3-17}$$

(3-16) is Newton for the whole system: total mass accelerates, plus a term for the rod swinging relative to the cart. (3-17) is the rod about the pivot: $I + ml^2$ is the parallel-axis inertia, $ml\ddot{x}$ is the kick the rod feels when the cart accelerates, $mgl\theta$ is gravity tipping it further. **The cart's acceleration is the only handle on the angle.**

Example 3-6 on the next page sets $I = 0$ (a point mass on a massless rod). The lab pole is a rod, so keep I. For a uniform rod of length $2l$,

$$I = \tfrac{1}{12}m(2l)^2 = \tfrac{1}{3}ml^2, \qquad J \equiv I + ml^2 = \tfrac{4}{3}ml^2 = 5.353 \times 10^{-3}\ \mathrm{kg\,m^2}.$$

This matches the handout's $J = \frac{1}{3}mL^2$ about the end, with $L = 2l$.

3 Keeping the sines gives the model swing-up needs

The linear model only holds within about 10° of upright (Section 8). Swing-up crosses every angle, so the simulator needs the full version. Differentiate the centre-of-mass coordinates twice:

$$\ddot{x}_G = \ddot{x} + l\ddot{\theta}\cos\theta - l\dot{\theta}^2\sin\theta, \qquad \ddot{y}_G = -l\ddot{\theta}\sin\theta - l\dot{\theta}^2\cos\theta.$$

(3-10) and (3-11) then give the pivot forces outright:

$$H = m(\ddot{x} + l\ddot{\theta}\cos\theta - l\dot{\theta}^2\sin\theta), \qquad V = mg - ml(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta).$$

Insert H into the cart equation (3-12). Insert both into (3-9); the $\dot{\theta}^2$ terms cancel and $\sin^2 + \cos^2 = 1$ collects the $\ddot{\theta}$ terms:

$$(M+m)\,\ddot{x} + ml\cos\theta\,\ddot{\theta} - ml\sin\theta\,\dot{\theta}^2 = u \tag{N1}$$

$$J\,\ddot{\theta} + ml\cos\theta\,\ddot{x} = mgl\sin\theta \tag{N2}$$

The new term $ml\sin\theta\,\dot{\theta}^2$ is the centripetal pull of the spinning rod on the cart. Setting $\cos\theta = 1$, $\sin\theta = \theta$, $\dot{\theta}^2 = 0$ gives back (3-16) and (3-17).

4 The lab measures the angle the other way: substitute $\theta = -\alpha$

Both angles are zero upright and differ only in sign. Every derivative carries the sign along; cosine is even and sine is odd:

$$\theta = -\alpha \quad \dot\theta = -\dot\alpha \quad \ddot\theta = -\ddot\alpha \quad \sin\theta = -\sin\alpha \quad \cos\theta = \cos\alpha$$

In (N1), $\cos\theta\,\ddot\theta \to -\cos\alpha\,\ddot\alpha$ and $\sin\theta\,\dot\theta^2 \to -\sin\alpha\,\dot\alpha^2$. (N2) becomes $-J\ddot\alpha + ml\cos\alpha\,\ddot x = -mgl\sin\alpha$; multiply by -1:

$$(M+m)\,\ddot x - ml\cos\alpha\,\ddot\alpha + ml\sin\alpha\,\dot\alpha^2 = u \tag{L1}$$

$$J\,\ddot\alpha - ml\cos\alpha\,\ddot x = mgl\sin\alpha \tag{L2}$$

Only the coupling terms change sign; gravity keeps its sign, because the pole falls away from upright whichever direction is called positive. Three checks:

- Push the cart right ($\ddot x > 0$, pole at rest upright). (L2) gives $\ddot\alpha > 0$: the pole lags and tips toward $-x$, counter-clockwise.
- Lean the pole ($\alpha > 0$, cart held). (L2) gives $\ddot\alpha > 0$: it keeps falling.
- (L2) rearranged is $J\ddot\alpha - mgl\sin\alpha = ml\ddot x\cos\alpha$, the equation printed in the handout's Lyapunov step (p. 15). The flip agrees with the course's own model.

Deriving from scratch with $x_G = x - l\sin\alpha$, $y_G = l\cos\alpha$ gives the same two lines.

5 The motor is a voltage-to-force gain with built-in damping

Exercise A-3-9 models a DC motor with the armature inductance neglected. Kirchhoff on the armature and the torque law:

$$V_m = R_m i + K_m\omega_m, \qquad \tau_m = K_t i.$$

The gearbox turns the motor K_g times faster than the pinion, and the pinion of radius r_{mp} rolls along the track: $\omega_m = K_g\dot x / r_{mp}$. The pinion delivers torque $K_g\tau_m$, so the force on the cart is $F_c = K_g\tau_m / r_{mp}$. Eliminate i and ω_m:

$$F_c = \frac{K_g K_t}{R_m r_{mp}}\Big(V_m - \frac{K_g K_m}{r_{mp}}\dot x\Big) = a\,V_m - b\,\dot x \tag{3}$$

which is the handout's equation (3), with

$a = K_g K_t/(R_m r_{mp})$	force per volt	1.724 N/V
$b = K_g^2 K_t K_m/(R_m r_{mp}^2)$	back-EMF damping	7.724 N s/m
$a \cdot 10\,\mathrm{V}$	force at standstill, full voltage	17.2 N
$a \cdot 10\,\mathrm{V}/b$	top speed, no friction	2.23 m/s

The back-EMF acts like friction you get for free: b adds to whatever B_f Step 2 measures. The net force on the cart, which replaces Ogata's u, is

$$F = aV_m - (b + B_f)\,\dot x.$$

Equation (3) also drops the rotor inertia. Reflected through the gear it behaves like $J_m K_g^2 / r_{mp}^2 = 0.133\,\mathrm{kg}$ of extra cart mass, 23 % of M. If the 10 Hz test in Section 8 shows a smaller swing on the rig than in simulation, add 0.133 to M in (L1) before blaming the friction.

6 Solved for the accelerations, the model is a simulator

(L1) and (L2) are linear in $\ddot{x}$ and $\ddot{\alpha}$. In matrix form, with $c = \cos\alpha$, $s = \sin\alpha$:

$$\begin{bmatrix} M+m & -mlc \\ -mlc & J \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\alpha} \end{bmatrix} = \begin{bmatrix} F - mls\,\dot{\alpha}^2 \\ mgl\,s \end{bmatrix}, \qquad D = (M+m)J - m^2l^2c^2.$$

$D > 0$ at every angle, since $(M+m)J > m^2l^2$. Inverting the 2×2 matrix:

$$\ddot{x} = \frac{J(F - mls\,\dot{\alpha}^2) + m^2l^2g\,s\,c}{D} \tag{S1}$$

$$\ddot{\alpha} = \frac{mlc(F - mls\,\dot{\alpha}^2) + (M+m)\,mgl\,s}{D} \tag{S2}$$

with $F = aV_m - (b+B_f)\dot{x}$. With state $[x_c, \alpha, \dot{x}_c, \dot{\alpha}]$, these two lines plus $\frac{d}{dt}x_c = \dot{x}_c$, $\frac{d}{dt}\alpha = \dot{\alpha}$ are the whole simulator. The appendix has it as MATLAB code (ode45 or a MATLAB Function block).

7 Linearised at upright: one integrator, one unstable pole

Set $c = 1$, $s = \alpha$, $\dot{\alpha}^2 = 0$ in the solved form. Then $D_0 = (M+m)J - m^2l^2 = 3.221 \times 10^{-3}$ and, with $x = [x_c, \alpha, \dot{x}_c, \dot{\alpha}]^\top$, $y = [x_c, \alpha]^\top$:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \dfrac{m^2l^2g}{D_0} & -\dfrac{J(b+B_f)}{D_0} & 0 \\ 0 & \dfrac{(M+m)mgl}{D_0} & -\dfrac{ml(b+B_f)}{D_0} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ Ja/D_0 \\ mla/D_0 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \qquad D = 0.$$

With the lab numbers and $B_f = 0$ until Step 2 supplies it:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1.553 & -12.84 & 0 \\ 0 & 47.93 & -54.14 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 0 \\ 2.864 \\ 12.08 \end{bmatrix}.$$

The third column scales as $-1.662\,(7.724 + B_f)$ and $-7.010\,(7.724 + B_f)$ once B_f is known. Both entries of B are positive: a positive voltage pushes the cart toward $+x$ and tips the pole counter-clockwise, as in Section 4.

The characteristic polynomial is $s\,(s^3 + 12.84s^2 - 47.93s - 531.1)$, with eigenvalues

0	cart position: nothing pulls the cart back to the centre
+6.60	the pole falling; its tilt grows by e every 0.15 s
−5.97	the mirror of the falling mode
−13.47	cart velocity, braked mostly by the back-EMF

Without any damping the pair would be $\pm\sqrt{47.93} = \pm 6.92$; the back-EMF slows the fall slightly. rank ctrb$(A, B) = 4$ and rank obsv$(A, C) = 4$: one voltage reaches all four states, and measuring x_c and α is enough to reconstruct the velocities. **Every closed-loop pole has to sit left of the $+6.6$ the pole brings, and the 10 V limit decides how far left**, which is why the handout's hint is $[-10, 0)$.

8 What the simulator predicts for the two tests

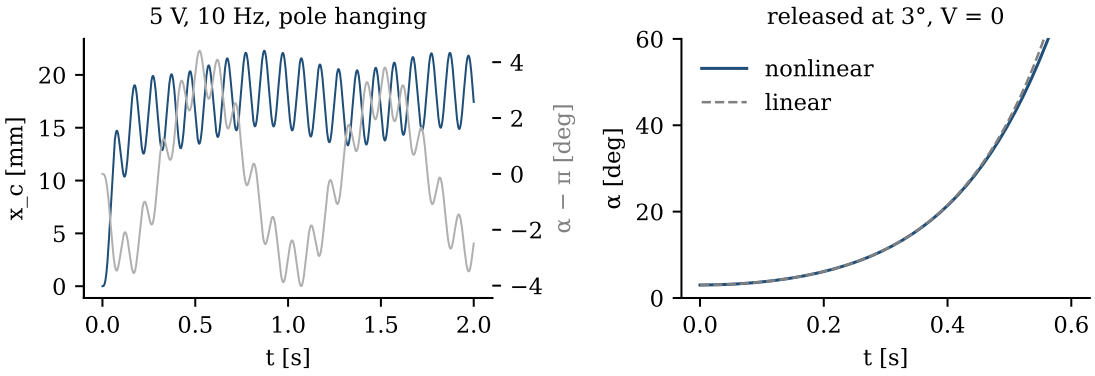


Figure 2. Left: Task 1's input, $5\sin(2\pi \cdot 10\,t)$ V, pole hanging ($\alpha = \pi$), cart in the centre. Right: pole released at 3° from upright with $V_m = 0$; nonlinear (L1, L2) against the linear $\dot{x} = Ax$. Both with $B_f = 0$.

The 10 Hz test should show the cart shaking 7 mm peak-to-peak around a point about 18 mm from where it started. The offset comes from the sine starting at zero phase: the first half-cycle pushes one way only, the cart picks up net momentum, and with no spring only friction stops it. The pole follows the 10 Hz shake by only 1°, but that first push also sets it swaying at its own frequency, about 1.1 Hz, by ±3.5°; with no pole friction in the model that sway never dies out. These numbers are what to read off the rig: the 10 Hz swing, the offset, the slow sway. A smaller 10 Hz swing on the hardware points to the missing 0.133 kg rotor mass; a smaller offset points to friction, which Step 2 measures; a sway that dies out shows friction in the pole bearing, which this model leaves out.

Released at 3°, the pole reaches 10° after 0.28 s and 30° after 0.45 s. At 10° the linear model still agrees to 0.1°; past that the two curves part. The LQR from Step 4 can trust its model within about 10°, and the handout's 5° switch-on window sits well inside that.

9 Step 2 falls out of (L1) for free

Put the friction back into (L1) explicitly and move everything known to one side:

$$\underbrace{F_c - (M+m)\ddot{x}_c + ml\big(\cos\alpha\,\ddot{\alpha} - \sin\alpha\,\dot{\alpha}^2\big)}_{y} = B_f\,\dot{x}_c.$$

So $\theta = B_f$ and $d = 1$. `regression.slx` already logs F_c (computed by equation (3)), x, $\dot{x}$, $\ddot{x}$, α, $\dot{\alpha}$, $\ddot{\alpha}$ to the workspace, and the least-squares estimate (6) collapses to a ratio of two dot products: $\hat{B}_f = \dot{x}^\top y / \dot{x}^\top \dot{x}$.

Three things to carry into the lab

(a) The model is (L1) and (L2) with $u = aV_m - (b + B_f)\dot{x}$. Simulator, regression and linearisation are rearrangements of those two lines.

(b) The angle convention only flips the coupling terms. If a controller pushes the pole the wrong way, check that sign first.

(c) The open-loop pole at $+6.6\,\text{rad/s}$ sets the timescale. Anything that reacts slower than about $0.15\,\text{s}$ loses the pole.

Appendix: the simulator in MATLAB

Variable names match the handout's setup script, so run `system_setup.m` first. Save the function below as `cartpole_f.m` in the same folder.

```
function ds = cartpole_f(t, s, V, P)
% s = [x_c; alpha; x_c_dot; alpha_dot], alpha = 0 upright, CCW positive
al = s(2); xd = s(3); w = s(4);
m = P.M_p; M = P.M_c; l = P.r_p; J = 4/3*m*l^2;
a = P.K_g*P.K_t/(P.R_m*P.r_mp);              % 1.724 N/V
b = P.K_g^2*P.K_t*P.K_m/(P.R_m*P.r_mp^2);    % 7.724 N s/m
F = a*V(t) - (b + P.B_f)*xd;                 % net force on the cart
c = cos(al); sn = sin(al);
D = (M+m)*J - (m*l*c)^2;
xdd = (J*(F - m*l*sn*w^2) + m^2*l^2*P.g*sn*c) / D;
add = (m*l*c*(F - m*l*sn*w^2) + (M+m)*m*P.g*l*sn) / D;
ds = [xd; w; xdd; add];
end
```

Task 1, the 10 Hz test from the hanging position:

```
system_setup;
P = struct('M_p',M_p,'M_c',M_c,'r_p',r_p,'g',g,'K_g',K_g,'K_t',K_t, ...
           'K_m',K_m,'R_m',R_m,'r_mp',r_mp,'B_f',0);
V = @(t) 5*sin(2*pi*10*t);
[t, s] = ode45(@(t,s) cartpole_f(t,s,V,P), [0 2], [0; pi; 0; 0], ...
               odeset('MaxStep',1e-3));
subplot(2,1,1); plot(t, 1e3*s(:,1)); ylabel('x_c [mm]');
subplot(2,1,2); plot(t, rad2deg(s(:,2)-pi)); ylabel('\alpha - \pi [deg]');
```

Step 4's matrices and the two rank checks:

```
m = M_p; M = M_c; l = r_p; J = 4/3*m*l^2; B_f = 0;
a = K_g*K_t/(R_m*r_mp); b = K_g^2*K_t*K_m/(R_m*r_mp^2);
D0 = (M+m)*J - m^2*l^2;
A = [0 0 1 0; 0 0 0 1;
     0 m^2*l^2*g/D0     -J*(b+B_f)/D0   0;
     0 (M+m)*m*g*l/D0   -m*l*(b+B_f)/D0 0];
B = [0; 0; J*a/D0; m*l*a/D0];
C = [1 0 0 0; 0 1 0 0]; D = [0; 0];
eig(A)                                   % 0, 6.60, -5.97, -13.47
rank(ctrb(A,B)), rank(obsv(A,C))         % 4, 4
```

In Simulink the same function goes into a MATLAB Function block with inputs V_m and the state, followed by four integrators fed back as the state. The initial condition sits on the integrators ($\alpha_0 = \pi$ for hanging).